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Odd-parity magnetism in non-collinear 2D materials

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1 Introduction

The subject of magnetism has experienced renewed interest due to the emergence of novel classes of magnetic materials [1]. Five years ago, a new type of magnetism, altermagnetism, was identified [2]. Altermagnetism integrates fundamental characteristics of two well-established classes of magnetism, ferromagnetism and antiferromagnetism. Analogously to ferromagnets, altermagnets exhibit non-relativistic spin-split electronic bands, while, like antiferromagnets, they possess a fully compensated magnetic structure resulting in zero net magnetization [2]. The collinear magnetic structure of altermagnets gives rise to an even-parity band dispersion [2].

In contrast, a less extensively explored class of unconventional magnets is the odd-parity, or p -wave, magnet, which can emerge from non-collinear magnetic configurations [1, 3]. These unconventional magnets have attracted growing interest within the scientific community, driven both by the pursuit of a deeper fundamental understanding and by their promising potential for quantum applications, such as in the field of spintronics [4], as well as the possibility of coexistence between odd-parity magnetism and superconductivity[5].

The search for odd-parity magnetic materials remains in its early stages, making Density Functional Theory (DFT) the natural playground for expanding the search for candidate materials. DFT is the most widely used method for calculating the electronic structure of materials, and its accuracy, versatility, and computational efficiency has made it central to materials science, design and discovery, ushering in the field of computational materials science. [6, 7]. At its core, DFT rests on the Hohenberg–Kohn theorems, which establish that the ground-state properties of a many-electron system are uniquely determined by the electron density. The electron density is then found self-consistently through the Kohn-Sham equations, which model the many-body interacting problem via a fictitious independent system constructed to have the same ground state electron density. Through this simplification, DFT becomes a natural and efficient tool for computing electronic band structures and screening candidate materials.

A central aspect in the study of unconventional magnetic materials is spin–orbit coupling (SOC), a relativistic interaction that couples the electron’s spin degree of freedom to its orbital motion. The strength of SOC scales approximately as Z^4 with the atomic number Z , implying that heavier elements exhibit substantially stronger spin–orbit interactions [8]. In unconventional magnets, spin splitting originates from non-relativistic exchange interactions, whereupon SOC may induce a distinct relativistic contribution. The competition and interplay between these relativistic and non-relativistic spin-splitting mechanisms constitute a compelling subject for further investigation [9].

In this project, we will use DFT to further explore the odd-parity properties of the spin textures in materials known from previous studies [10] to host coplanar spiral magnetic order. We will analyze the underlying symmetries and perform first-principles calculations of the electronic band structure to verify that the spin texture exhibits the predicted symmetry in k -space. The primary aim of this project will then be to investigate how the inclusion of SOC alters this odd-parity spin texture, as SOC may introduce competing magnetic interactions, inducing anisotropy in the magnetic order. This project is structured as follows: In Section 2, we will go through the essential theoretical background of DFT and odd-parity magnetism. In Section 3, we introduce the materials of interest and the DFT method used to examine them in this project. In Section 4 we will present the results from the above mentioned search and discuss how SOC alters the magnetic structure, with a final summarizing conclusion in Section 5.

2 Theory

In this section we will first go through density functional theory (DFT) and why it is useful. Afterwards we end the section with a small introduction to p -wave magnets and their symmetries.

2.1 Density Functional Theory

To develop a better understanding of DFT, we first review the essential concepts required to describe electronic properties. This will ultimately allow us examine the effects of SOC in odd-parity magnets. The following section is based primarily on Chapters 3 and 9 from "Materials modelling using Density Functional Theory" by Feliciano Giustino [11].

2.1.1 Ground state energy

In a quantum mechanical system, one may consider an arbitrary number, N , electrons, and the corresponding (nonrelativistic) many-electron wavefunction depends on $3N$ Cartesian spatial coordinates. Consequently, the computational cost of treating systems with large N grows rapidly, rendering direct calculations increasingly demanding. This complexity arises even when the nuclei are treated as stationary (the Born-Oppenheimer approximation), and before explicitly incorporating electron–electron interactions or other many-body effects. From quantum mechanics we know that we can write the total energy of a Hamiltonian as,

$$E = \langle \Psi | H | \Psi \rangle = \int d\mathbf{r}_1 \dots d\mathbf{r}_N \Psi^*(\mathbf{r}_1 \dots \mathbf{r}_N) H \Psi(\mathbf{r}_1 \dots \mathbf{r}_N), \quad (1)$$

$$E = \mathcal{F}[\Psi], \quad (2)$$

where the last line is written to indicate that the energy is a functional of the many-body wavefunction. The main idea of DFT is that we can derive the ground state energy as functional of the ground state density, which is a function of only 3 spatial coordinates. This is the remarkable discovery by Hohenberg and Kohn (1964) [12]. The Hohenberg–Kohn theorem relies on three fundamental premises. First, the electron density in the ground state uniquely determines the external potential generated by the nuclei. Second, this external potential uniquely determines the many-body wavefunction. The third and final premise is that the total energy of the system can be expressed as a functional of the many-body wavefunction. By combining the premises, the ground state energy can be uniquely connected with the ground state density, and hence we can rewrite Eq. 2 to $E = \mathcal{F}[n]$. The next step consists of constructing a fictitious system of independent electrons that reproduces the properties of the real system described in Eq. 1 with the highest possible accuracy. The fictitious system is known as the

Kohn-Sham system.

To obtain the energy as a functional of the electronic density, we begin by expressing the general many-electron Hamiltonian as the sum of a kinetic-energy operator, T , an external potential term, V_n , and the electron–electron Coulomb interaction, W .

$$H(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = T + \sum_i V_n(\mathbf{r}_i) + W, \quad (3)$$

$$T = - \sum_i \frac{1}{2} \nabla_i^2, \quad W = \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (4)$$

where the nuclei are fixed so their kinetic energy vanishes, and their mutual Coulomb repulsion enters as a constant energy shift in the external potential. The energy can be calculated,

$$E = \langle \Psi | \sum_i V_n(\mathbf{r}_i) | \Psi \rangle + \langle \Psi | T + W | \Psi \rangle, \quad (5)$$

$$= \sum_i \int d\mathbf{r}_1 \dots d\mathbf{r}_N |\phi|^2 V_n(\mathbf{r}_i) + \langle \Psi | T + W | \Psi \rangle, \quad (6)$$

$$= \int d\mathbf{r} n(\mathbf{r}) V_n(\mathbf{r}) - \sum_i \int d\mathbf{r} \phi_i^*(\mathbf{r}) \frac{\nabla^2}{2} \phi_i(\mathbf{r}) + \frac{1}{2} \int \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + E_{xc}[n], \quad (7)$$

where the first term is the external potential expressed using the electronic density. Here the $\phi_i(\mathbf{r})$ are the single-particle wavefunctions. The second and third term is the kinetic and Coulomb energy of the independent electrons, and the fourth term is the exchange-correlation term, which accounts for the discrepancy between the energy of the independent electrons and the energy of the real system in Eq. 1. It is thus evident that the energy in Eq. 7 should now be regarded as a functional of the density. In the end, determining the exchange-correlation energy is the tricky part, and several different methods exist to approximate it. By using the variational principle and requiring the Kohn-Sham wavefunctions ϕ_i be orthonormal, Eq. 7 can be turned into,

$$\left[-\frac{1}{2} \nabla^2 + V_n(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc} \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (8)$$

$$V_H(\mathbf{r}) = \int d\mathbf{r}' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad n(\mathbf{r}) = \sum_i |\phi_i(\mathbf{r})|^2, \quad (9)$$

which are known as the Kohn–Sham equations with eigenvector $\phi_i(\mathbf{r})$ and eigenvalues ε_i . Here, the ground-state density n is defined as the density that minimizes the ground-state energy functional. This procedure leads to a self-consistent process: one begins with an initial guess for the electron density, solves the Kohn–Sham equations to obtain the corresponding eigenvalues and eigenvectors, and then reconstructs a new density from these eigenvectors. The process is iterated until the resulting

density converges, that is, the variation is below some defined threshold.

2.1.2 Band structure

In this project, we aim to investigate the electronic properties of materials by analysing their band structures. Consequently, it is crucial to understand how the above mentioned Kohn–Sham equations are related to the band structure of a given material. It is important to emphasize that the Kohn–Sham eigenvectors do not correspond to the true many-body wavefunctions of a real material. Rather, they are a mathematical construct within the Kohn–Sham formalism. In many-cases, this fictitious system is a good estimate of the real system, and enough to understand many electronic and optical properties. By rewriting the energy in terms of occupation,

$$E = \sum_i f_i \varepsilon_i - \left[E_H + \int d\mathbf{r} V_{xc}(\mathbf{r}) n(\mathbf{r}) - E_{xc} \right], \quad (10)$$

with $f_i = 1$ for occupied states and 0 for unoccupied states. Since the band structure encodes the quasi-particle excitation spectrum, it is natural to connect the Kohn–Sham eigenvalues and the total energy of the system. To this end, we consider the change in the total energy upon the addition of a single electron and analyze the corresponding energy difference before and after this electron is introduced,

$$E_{N+1,i} - E_N = \int_0^1 df_i \frac{\partial E}{\partial f_i} = \int_0^1 df_i \varepsilon_i, \quad (11)$$

$$\simeq \varepsilon_i. \quad (12)$$

Here we have used that, in large solids, adding one electron does not change the density significantly along with the fact that only the first term in Eq. 10 depends on the occupation. Adding an electron to the system on the right-hand side of Eq. 11 corresponds to the occupied eigenvalue of the Kohn–Sham equations. In crystalline solids we express the wavefunctions via Bloch’s theorem, $\phi_{i\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} u_{i\mathbf{k}}(\mathbf{r})$, which is a product of a plane wave and periodic function in the unit cell. This alters our energy expression from Eq. 2,

$$\left[-\frac{1}{2}(\nabla + i\mathbf{k})^2 + V_n(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc} \right] u_{i\mathbf{k}}(\mathbf{r}) = \varepsilon_{i\mathbf{k}} u_{i\mathbf{k}}(\mathbf{r}), \quad (13)$$

where we explicitly did the second derivative of the Bloch eigenstates. Further, this changes the density to

$$n(\mathbf{r}) = \sum_i \int_{BZ} \frac{d\mathbf{k}}{\Omega_{BZ}} f_{i\mathbf{k}} |u_{i\mathbf{k}}|^2, \quad (14)$$

here Ω_{BZ} is the volume of the Brillouin zone. With this notion of the density in crystallines it is possible to rewrite the total energy from Eq. 10 to the total energy of a crystalline structure,

$$E = \sum_i \int_{BZ} \frac{d\mathbf{k}}{\Omega_{BZ}} f_{i\mathbf{k}} \varepsilon_i - \left[E_H + \int d\mathbf{r} V_{xc}(\mathbf{r}) n(\mathbf{r}) - E_{xc} \right], \quad (15)$$

where together for the Eq. 14 this constitutes the self-consistent Kohn-Sham equations for crystalline solids. One should note that the Kohn-Sham equations are exactly the same as the energy of a real material if we know the exchange-correlation function exactly. But obtaining the exact formulation of the exchange-correlation energy is exactly the tricky part. It is therefore approximated, in the simplest case with the local density approximation (LDA), which depends solely on the electronic density at each point in space. It is therefore important to remember that the band structures that we will show in the rest of the project are solutions to the Kohn-Sham equations.

2.2 Odd-parity magnets

In the field of magnetism the importance of symmetries has long been realized [13]. Here, the preservation or violation of inversion (P) and time-reversal (TR) symmetry is of special interest, as they, respectively, enforce the degeneracy of $E_{\uparrow}(\mathbf{k}) = E_{\uparrow}(-\mathbf{k})$ and $E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(-\mathbf{k})$. These two together thus imply spin-bands being degenerate. However, as any magnetic order must break TR, a so called effective TR is often defined. This combines TR with another symmetry operation, such as translation or rotation, which together leave the system invariant, thus enforcing degeneracy points for energies with opposite spin. As an example, the defining feature of the anti-ferromagnetic order is the preservation of P and the effective TR consisting of TR and a translation (T). With T acting trivially in momentum space, this enforces $E_{\uparrow}(\mathbf{k}) = E_{\uparrow}(-\mathbf{k}) = E_{\downarrow}(-\mathbf{k}) = E_{\downarrow}(\mathbf{k})$, giving rise to an effective Kramers degeneracy, where bands with opposite spin are forced to be degenerate. An essential condition for materials to exhibit a non trivial spin texture is thus to lift the Kramers degeneracy.

In the case of odd-parity magnets, the lifting Kramers degeneracy is caused by the violation of P, while preserving some effective TR [14]. Here, systems hosting a coplanar spin spiral magnetic order is of special interest, as the magnetic order intrinsically violates P. However, with the magnetic order

being coplanar, spin is no longer a good quantum number, and we can therefore only talk about the spin expectation value $\langle s_i \rangle$ for $i \in \{x, y, z\}$. Using the symmetry of the system, we can investigate the allowed values of $\langle s_i \rangle$. Consider a spin spiral consisting of N_s magnetic sites that, in the absence of magnetic order, are related by a translation τ . In the spin-spiral state, the magnetic moments on neighbouring sites differ by a relative in-plane angle $\theta = 2\pi/N_s$. The system then has the combined symmetry of $C_{\theta\hat{z}}$ with τ , where $C_{\theta\hat{z}}$ is a rotation of the magnetic moments by the angle θ around the spin-spiral normal vector (chosen here to be $\hat{z}$), and τ is the translation. This symmetry operation acts on the spin expectation values in the following way:

$$C_{\theta\hat{z}}\tau\mathbf{S}(\mathbf{k}) = C_{\theta\hat{z}}\tau \begin{pmatrix} \langle s_x(\mathbf{k}) \rangle \\ \langle s_y(\mathbf{k}) \rangle \\ \langle s_z(\mathbf{k}) \rangle \end{pmatrix} = C_{\theta\hat{z}} \begin{pmatrix} \langle s_x(\mathbf{k}) \rangle \\ \langle s_y(\mathbf{k}) \rangle \\ \langle s_z(\mathbf{k}) \rangle \end{pmatrix} = C_{\theta\hat{z}} \begin{pmatrix} \cos\theta\langle s_x(\mathbf{k}) \rangle - \sin\theta\langle s_y(\mathbf{k}) \rangle \\ \cos\theta\langle s_y(\mathbf{k}) \rangle + \sin\theta\langle s_x(\mathbf{k}) \rangle \\ \langle s_z(\mathbf{k}) \rangle \end{pmatrix}. \quad (16)$$

However, since this operation leaves the system invariant, it must be true that $\mathbf{S}(\mathbf{k}) = C_{\theta\hat{z}}\tau\mathbf{S}(\mathbf{k})$. This directly implies $\langle s_x(\mathbf{k}) \rangle = \langle s_y(\mathbf{k}) \rangle = 0$. We can further define the effective TR of the system, combining TR and a π rotation of the magnetic moments around $\hat{z}$, $C_{\pi\hat{z}}$. Letting this act on $\langle s_z(\mathbf{k}) \rangle$:

$$\langle s_z(\mathbf{k}) \rangle = \text{TR } C_{\pi\hat{z}} \langle s_z(\mathbf{k}) \rangle = \text{TR} \langle s_z(\mathbf{k}) \rangle = \langle -s_z(-\mathbf{k}) \rangle, \quad (17)$$

meaning, any expectation value in z must be odd in $\mathbf{k}$ [14]. From this symmetry analysis, we conclude that systems hosting a coplanar spin-spiral order have a vanishing expectation value for the in-plane electron spin components. In contrast, the expectation value of the out-of-plane spin component may be finite, but must then be odd in momentum. As a consequence, the electronic spin polarization acquires a nontrivial momentum dependence in the expectation value of the spin perpendicular to the underlying magnetic order. Materials exhibiting such behaviour have therefore been termed odd-parity magnets.

However, the inclusion of SOC induces a small out-of-plane magnetic component, such that the magnetic order can no longer be regarded as strictly coplanar. Consequently, the symmetry analysis presented above no longer strictly applies. Furthermore, in the presence of SOC, spin and spatial degrees of freedom can no longer be transformed independently, as they become intrinsically coupled. Since SOC is typically much weaker than the magnetic energy scale, an important question is therefore to what extent the odd-parity spin structure remains robust in the presence of SOC.

3 Method

In this section, we describe the criteria and procedure used for the selection of materials, followed by a detailed account of the construction of the magnetic unit cell. Finally, we present the density functional theory (DFT) computational setup employed in this work.

3.1 Materials

To identify the most suitable materials for investigation, we begin from the supplementary information of Ref. [10]. We restrict our search to hexagonal lattices exhibiting a spin spiral characterized by $\mathbf{Q} = [1/3, 1/3]$, with $\mathbf{Q}$ being the spiral wave-vector. The spin-spiral wave-vector defines the spatial periodicity and propagation direction of a non-collinear magnetic structure where atomic magnetic moments rotate in a helical fashion along the lattice.

In Table 1, we summarize all spin-spiral states found with this specific $\mathbf{Q}$ -vector. In this table, the angle θ denotes the polar angle of the spin-spiral normal vector with respect to the lattice: $\theta = 0$ indicates that the normal vector is oriented strictly out-of-plane, corresponding to an in-plane spin spiral. The angle ϕ specifies the in-plane azimuthal rotation.

Formula	$\mathbf{Q}$	SG	(θ, ϕ)	Formula	$\mathbf{Q}$	SG	(θ, ϕ)
Ag ₂ ReBr ₆	[1/3, 1/3]	$P\bar{3}1m$	(86, 190)	HfNiF ₆	-	-	(86, 94)
Ag ₂ ReCl ₆	-	-	(90, 60)	HfPdF ₆	-	-	(0, 0)
Ag ₂ ReF ₆	-	$P\bar{3}$	(90, 38)	In ₂ MnS ₄	-	$P\bar{3}m1$	(90, 266)
Ag ₂ ReI ₆	-	$P\bar{3}1m$	(22, 199)	MnBr ₂	-	$P\bar{3}m1$	(90, 236)
Al ₂ VS ₄	-	-	(2, 120)	MnCl ₂	-	-	(56, 319)
Al ₂ VSe ₄	-	-	(90, 242)	MnI ₂	-	-	(0, 0)
Bi ₂ MnS ₄	-	-	(90, 328)	MnTe ₂	-	$P\bar{6}m2$	(0, 0)
Bi ₂ MnSe ₄	-	-	(90, 240)	MnZrBr ₆	-	$P312$	(14, 254)
Bi ₂ MnTe ₄	-	-	(76, 265)	MnZrI ₆	-	-	(0, 0)
CrSe ₂	-	$P\bar{4}m2$	(90, 44)	NiH ₂ O ₂	-	$P\bar{3}m1$	(0, 0)
Cu ₂ ReI ₆	-	$P\bar{3}$	(0, 0)	VBr ₂	-	-	(88, 42)
Ga ₂ MnS ₄	-	$P3m1$	(40, 276)	VBrCl	-	$P3m1$	(2, 300)
Ga ₂ NiS ₄	-	$P\bar{3}m1$	(90, 332)	VBrI	-	-	(90, 148)
Ga ₂ VS ₄	-	-	(88, 148)	VBrI	-	-	(0, 0)
HfMnBr ₆	-	$P312$	(2, 120)	VCl ₂	-	$P\bar{3}m1$	(90, 58)
HfMnCl ₆	-	-	(76, 311)	VClI	-	$P3m1$	(90, 150)
HfMnI ₆	-	-	(0, 0)				

Table 1: Materials with $\mathbf{Q} = [1/3, 1/3]$ taken from Ref. [10]. SG refers to the space group of the material. (θ, ϕ) is the angle of spin-spiral normal vector and in-plane azimuthal rotation. The materials highlighted in red is metals, where purple is the control material.

The material highlighted in purple corresponds to the insulating control sample, which has also been examined in Ref. [15]. The materials highlighted in red are listed in C2DB [16] as metallic. Thus, while keeping $\mathbf{Q}$ and the lattice structure fixed, we examine materials with different $\hat{n}$ orientations,

space groups, and electronic properties.

3.2 Magnetic structure

To capture the spin-spiral effects, one needs to go beyond the primitive unit to the magnetic supercell, the minimum unit capturing the repetition of local magnetic moments. As we have restricted only to the case of $\mathbf{Q} = [1/3, 1/3]$ and hexagonal crystal structures, we can leverage the same rhombus shaped supercell chosen in Ref. [15]. We define a transformation matrix from the primitive to the magnetic cell as $P = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$. Figure 1(d) from Ref. [15] is reproduced in Fig 1 for convenience, illustrating the unit cell vectors, the primitive and the magnetic unit cells.

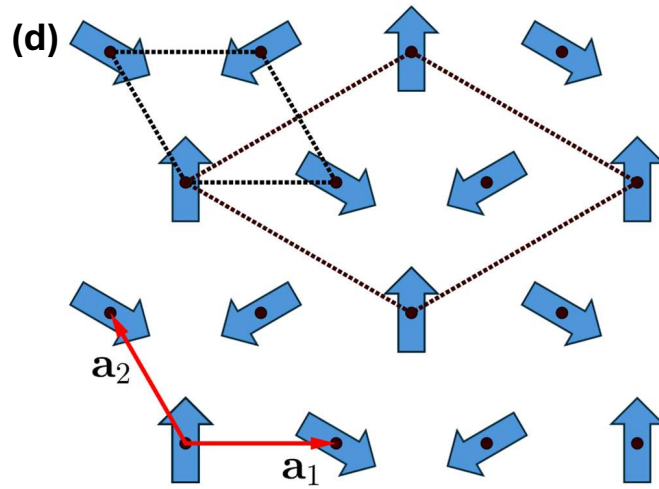


Figure 1: Figure 1(d) from Ref. [15]. Note that this corresponds to a counter-clockwise rotation if the normal vector points into the paper, $\hat{n} = -\hat{z}$.

The common convention for the spin-spiral orientation is counter-clockwise (CCW) around the normal vector. In all the following, we will denote the orientation of the spin-spiral normal vector as $\hat{n}$, with the understanding that the spiral is CCW around that vector. When SOC is not enabled, these orientations are energy degenerate. However, as we will be including SOC, a CW and CCW orientation of the spin spiral need not be energy degenerate. The degeneracy of chiral states is governed by the inversion symmetry of the crystal lattice. In an inversion-symmetric crystal, a clockwise configuration is transformed into a counter-clockwise configuration under spatial inversion, indicating that the two chiralities are strictly equivalent in energy. Conversely, in the absence of inversion symmetry, clockwise and counter-clockwise configurations are no longer related by inversion, and therefore are not necessarily energetically degenerate.

3.3 DFT setup

For the DFT self-consistent field (SCF) calculations, we use the DFT package GPAW [17] with the LDA exchange-correlation functional. LDA is chosen as, for non-coplanar magnetic order, it is the only

exchange-correlation function that is implemented correctly. A Fermi-smearing of 0.01eV was used for all calculations. To determine the input parameters, a convergence test for the MnI₂ SCF calculations was performed. Due to the spin-spiral structure, we enforce a non-collinear magnetic moment structure, and symmetry reduction was disabled in all calculations. Converging k -points and plane-wave (PW) cutoffs separately, we arrive at Figure 2. As convergence parameters, we examined the difference in energy per atom between the spin-spiral configuration and a ferromagnetic configuration. As an extra parameter, we include the magnetic moment magnitude on the Mn atoms.

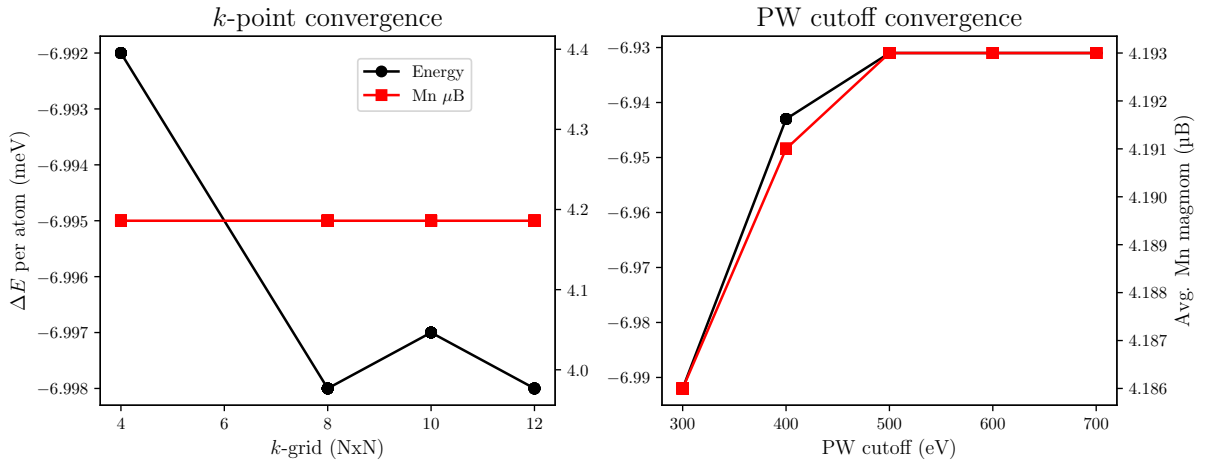


Figure 2: Convergence test results for MnI₂ without SOC. The black line shows the energy difference per primitive cell, the red shows the average magnetic moment magnitude on the Mn atoms. The PW cutoff is safely converged above 500eV, While the energy variation on the k -grid is very small for this material in general, with a difference between $4 \times 4 \times 1$ and $12 \times 12 \times 1$ being only 0.006meV. The magnetic moment magnitude is seen to be independent of the k -grid size.

As MnI₂ is an insulator, it may prove more stable than a metal [18]. Furthermore, SOC may affect the convergence. Hence, convergence tests for MnTe₂ with SOC enabled were also performed. Since the crystal structure of MnTe₂ lacks inversion, the energy comparison is taken to be the difference between the two different spin-spiral chiralities, as these are no longer degenerate (see Section 4.2). The result can be seen in Figure 3.

We aim to keep the numerical uncertainty below about 1% on the relevant energy scale. Since the total energy difference per primitive cell is of the order $\sim 23 - 26$ meV, we choose that numerical variations should be below about 0.3meV. In Figure 3, the difference between the k -grid sizes $10 \times 10 \times 1$ and $12 \times 12 \times 1$ is 0.27meV, so $12 \times 12 \times 1$ is utilized for all subsequent calculations. In both Figure 2 and Figure 3 the PW cutoff is safely converged beyond 500eV, so a cut of 600eV was used to be on the safe side.

For all the calculations we used the default GPAW convergence criteria of a maximum total energy

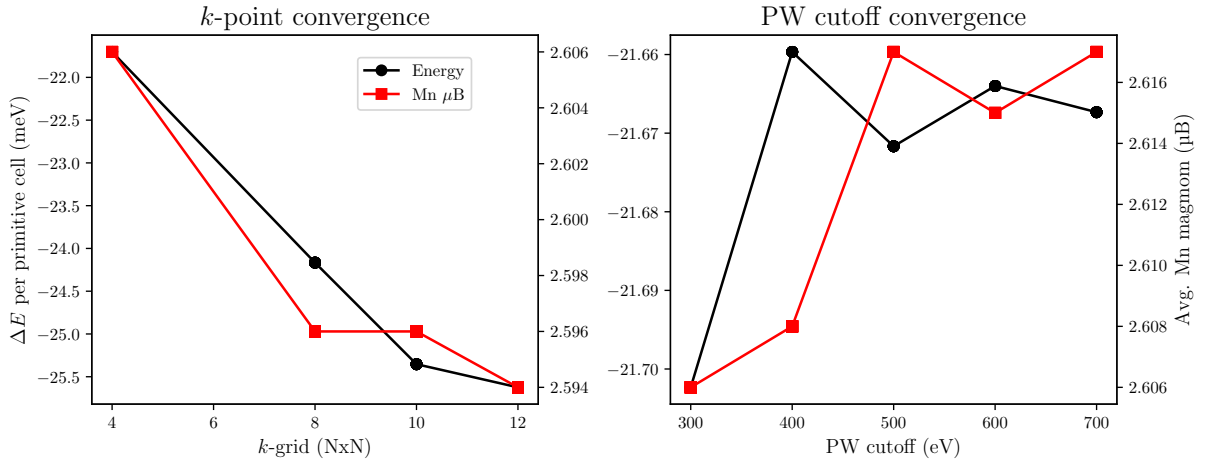


Figure 3: Convergence test results for MnTe₂ with SOC. The black line shows the energy difference $\Delta E = E(\hat{n} = \hat{z}) - E(\hat{n} = -\hat{z})$ per primitive cell, the red shows the average magnetic moment magnitude on the Mn atoms.

change in the last 3 cycles of 0.0005eV per valence electron, and a maximum integral of absolute eigenstate change of $4 \cdot 10^{-8} \text{eV}^2$ per valence electron, along with a maximum integral of absolute density change of 0.0001 electrons per valence electron. As an extra verification, due to small numerical fluctuations in the local magnetic moments when SOC is enabled, we ran the SOC calculation for MnI₂ with a convergence criterium on the density of 10^{-9} electrons per valence electron. We found that the spin-spiral structure was conserved with SOC enabled, with very small local moments accumulating on the iodine atoms.

After SCF calculation, we ran fixed density band structure calculations. It is important to note that these bands are in the supercell, not the primitive unit cell.

4 Results

In this section we will employ the DFT method described above for different two-dimensional materials having the odd-parity structure. We start by examining MnI_2 in detail, before moving on to examine the metals MnTe_2 and Al_2VSe_4 . The atomic structures are provided by C2DB [16]. For each material, we calculate the spin-resolved band structure on the high symmetry path $-M \rightarrow \Gamma \rightarrow M$ in the magnetic Brillouin zone, both with and without SOC. We also examine the atomic- and orbital projections of each material. For MnI_2 in particular, serving as our reference material, we also examine the case of SOC with stricter convergence criteria, and we examine a spin-spiral structure with the normal vector tilted out of the SOC-preferred orientation. All data and code for these calculations are publicly available on GitHub [19].

4.1 MnI_2

Before calculating the SCF with SOC enabled, we examine the preferred orientation of the spin-spiral normal vector $\hat{n}$. As MnI_2 is inversion-symmetric without the spin-spiral, we expect that the chiralities should be degenerate. This is done by including SOC non-self-consistently, and examining the energy cost of different orientations of the spiral. The result can be seen in Figure 4.

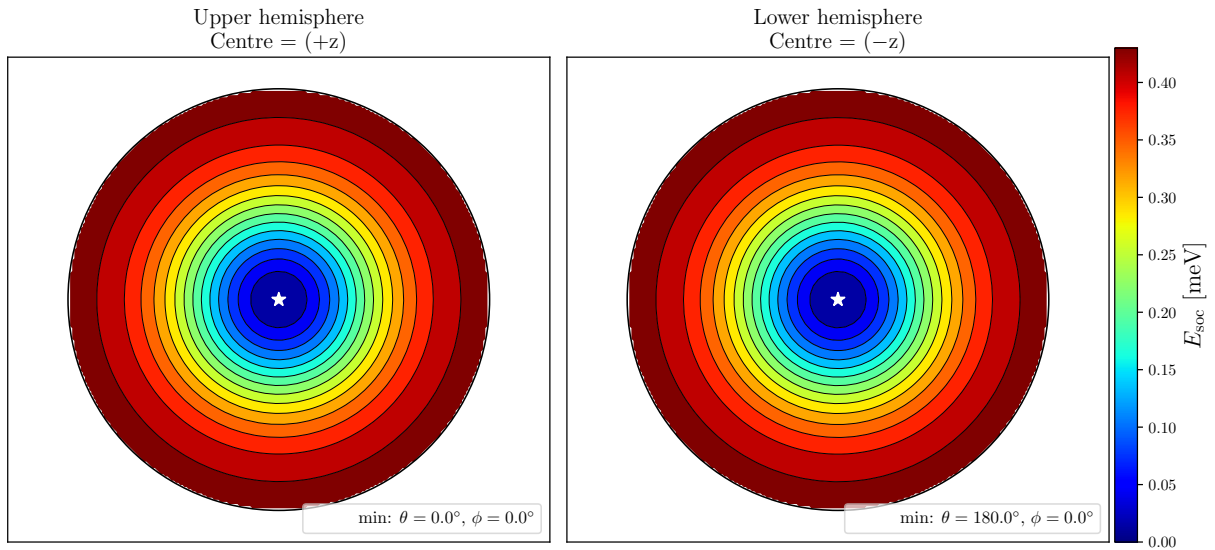


Figure 4: Energy cost of different orientations of $\hat{n}$ for MnI_2 , given in polar angle coordinates. The energy scale goes from the global minimum to the global maximum. The minimum is when $\hat{n}$ points directly out of plane, with the $+\hat{z}$ and $-\hat{z}$ directions being degenerate, confirming that chirality is not a factor for this material.

For MnI_2 we find no effect of chirality. The energy is identical for the normal vector in the $+z$ or $-z$ direction, and thus clockwise and counter-clockwise spirals are degenerate. We therefore proceed to perform a SCF calculation with SOC enabled with a clockwise rotating spiral. We include the

effect of SOC with an out-of-plane normal vector on the odd-parity magnetic structure and establish a benchmark. We repeated the SOC SCF and band structure calculations for a spin-spiral with a normal vector in the $-\hat{x}$ direction. This is not the preferred orientation of the spiral, but is included to explore how SOC oriented parallel with the normal vector alters the odd-parity spin texture. As all orientations of the spin-spiral are degenerate without SOC, it is sufficient to look only at the band structure when SOC is enabled in this case. Note that the spin expectation value is in the normal-vector direction.

The resulting band structure, compared to the case without SOC, can be seen in Figure 5. In the left side of the figure, the odd-parity structure is significant for the bands around -2eV . Including SOC and $\hat{n} = -\hat{z}$ in the middle of the figure, we see the quadruple crossing at Γ is split. The odd-parity structure is still visible at Γ for energies around -1.5eV , though the magnitude of the spin expectation value is smaller. By changing the spin-spiral normal vector to the x -direction as seen on the right in Figure 5 the odd-parity structure is more pronounced.

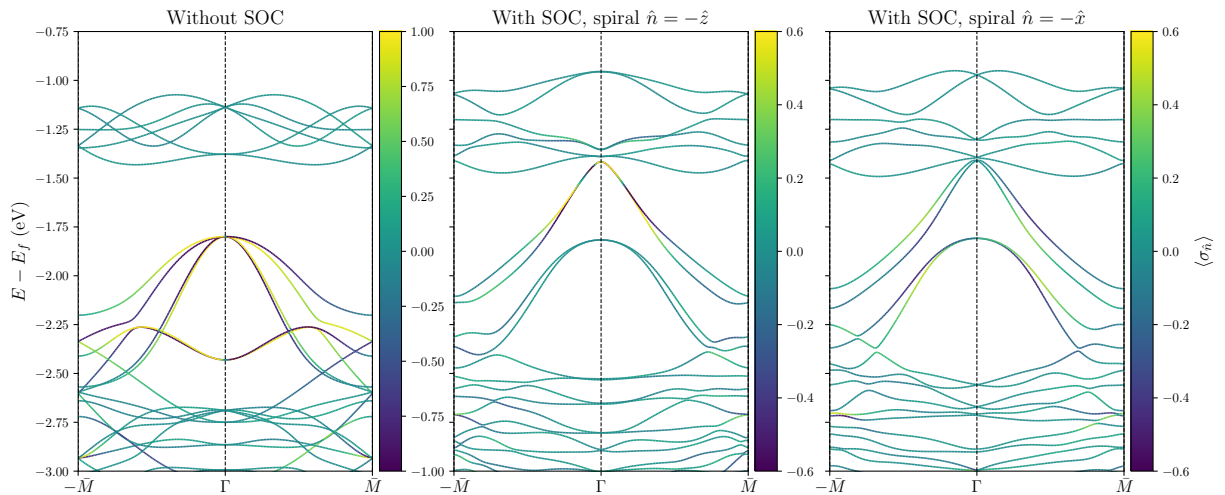


Figure 5: Spin-resolved band structure for MnI_2 for (left) without SOC (middle) with SOC and $\hat{n} = -\hat{z}$ (right) with SOC and $\hat{n} = -\hat{x}$. The bands are colored according to $\langle\sigma_n\rangle$. Note the different scale with and without SOC. All dispersions are plotted along the path $-\bar{M} - \Gamma - \bar{M}$ in the magnetic Brillouin zone.

4.1.1 Atomic- and Orbital Projections

To understand the nature of the odd-parity spin structure of some of the bands, a projection onto atomic orbitals was performed. GPAWs internal projector structure allows for easy extraction of the weights on both the atoms and the individual orbitals. It should be noted that the bases employed for both the atomic and orbital decompositions do not constitute complete sets and, consequently, the corresponding contributions do not sum to unity.

In Figure 6 we show in the upper row manganese decomposition into atomic, p -orbital and d -orbital respectively. In the lower panel the same decomposition for iodine is shown. In the first column it becomes evident that the odd-parity bands are mostly iodine. From the orbital decomposition, in the second and third column, it becomes evident that the main contribution to the odd-parity structure is from the p -orbitals of iodine. Furthermore, we find that the weight on the bands with odd-parity structure on manganese is also primarily in the p -orbitals.

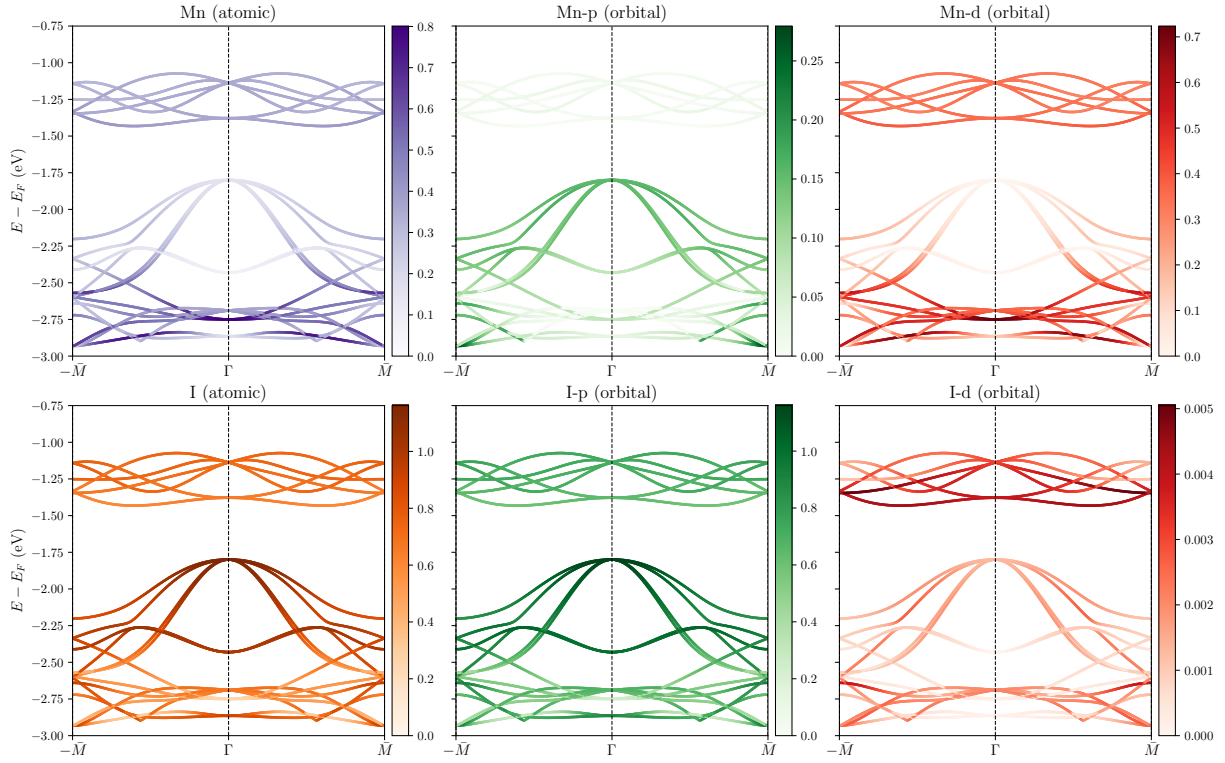


Figure 6: The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows iodine decomposition into atomic, p -orbital and d -orbitals respectively. Note that the weights need not sum to 1.

The same calculation was done with SOC enabled for both direction of the normal-vector, to see if the weights change. There is no significant change, the results are included in Appendix A.1.

4.2 MnTe₂

Again, before running a full SCF calculation with spin-orbit coupling, we examine the preferred orientation of $\hat{n}$ non-self-consistently. As MnTe₂ is not inversion-symmetric even without a spin-spiral, it is possible that chirality may be of import. The results are seen in Figure 7.

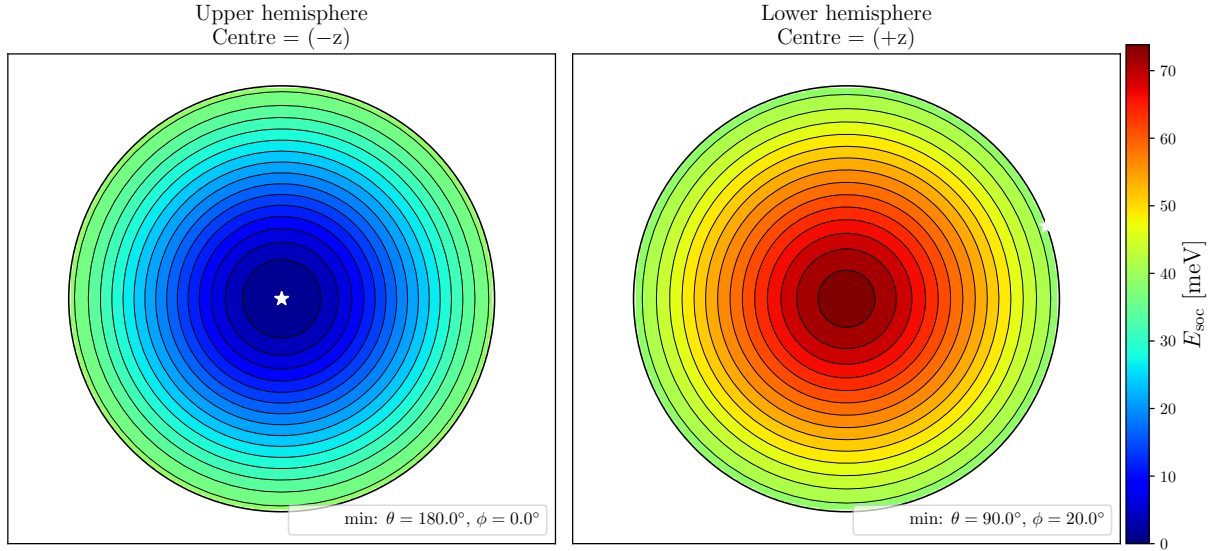


Figure 7: Energy cost of different orientations of $\hat{n}$ for MnTe₂, given in polar angle coordinates. The energy scale goes from the global minimum to the global maximum. Chirality is seen to be a major factor for this material.

For MnTe₂, seeing that chirality may play a major role, we ran the calculations with SOC for both a clockwise- and counter-clockwise spin spiral, see the resulting band structure in Figure 8. The two spiral orientations show remarkably different behaviours. We found that the CCW spin spiral was the ground state, with a direct gap of 0.088eV at Γ . The CW spiral, on the other hand, is metallic, with a Fermi energy of -0.403eV . The energy difference in the supercell between the two states is 76.7meV (meaning 25.2meV per primitive cell), aligning with the expectation from the non-self-consistent calculation seen in Figure 7.

For this material, the odd-parity structure is present close to the Fermi surface for the metallic configuration and near the valence band maximum in the gaped configuration. In this gaped configuration, the odd-parity structure appears to be more robust to the inclusion of SOC. It may therefore prove more interesting for spintronics applications.

4.2.1 Atomic- and Orbital Projections

We did the same decomposition for MnTe₂ into atoms and orbitals. From the first column of Figure 9, it is difficult to unambiguously identify the orbital localization of the atomic species, as both man-

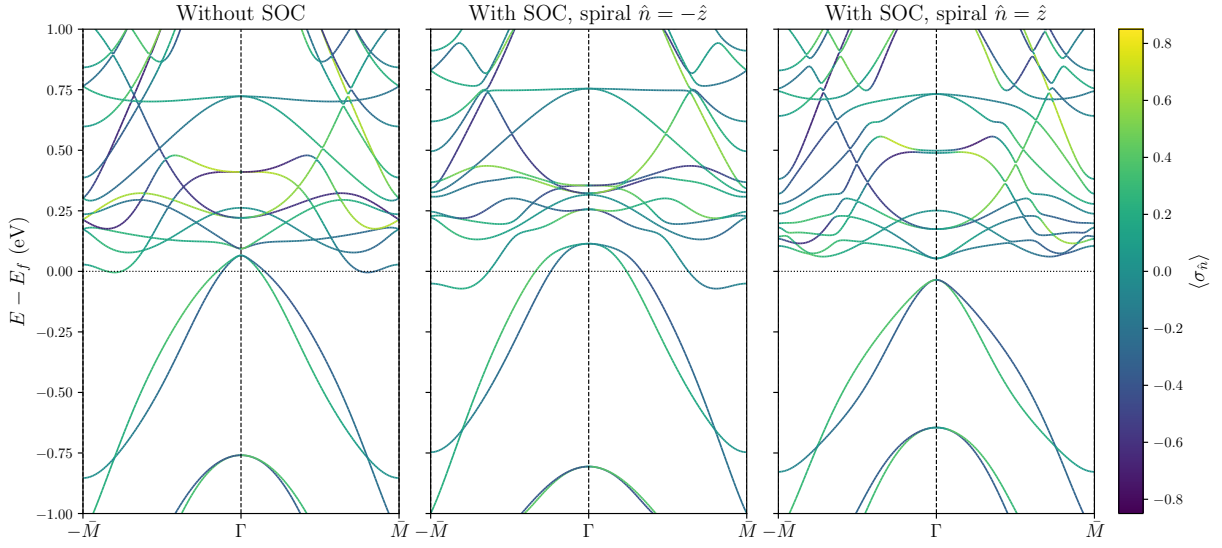


Figure 8: Spin-resolved band structure for MnTe_2 for (left) without SOC (middle) with SOC and $\hat{n} = -\hat{z}$ (right) with SOC and $\hat{n} = \hat{z}$. The bands are colored according to $\langle \sigma_n \rangle$. All dispersions are plotted along the path $-\bar{M} - \Gamma - \bar{M}$ in the magnetic Brillouin zone.

ganese and tellurium contributions are distributed across all bands. At the Fermi level, the manganese contribution is slightly larger than that of tellurium.

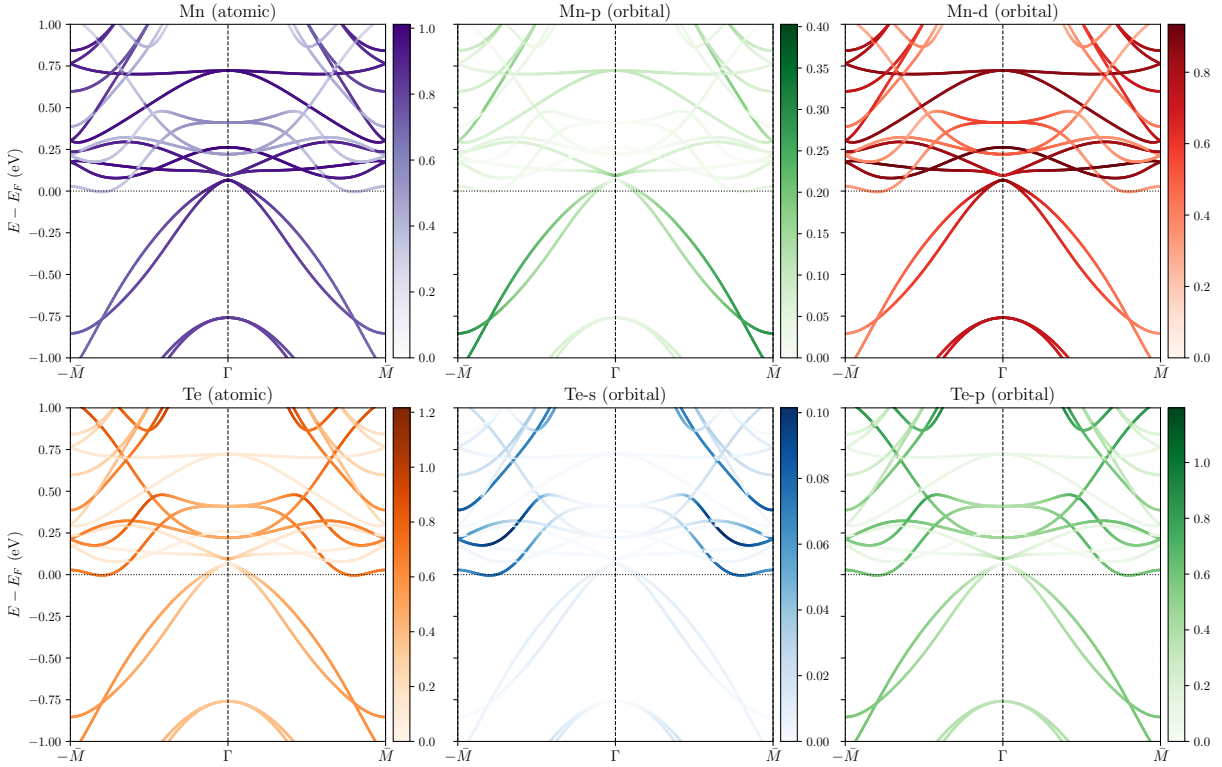


Figure 9: The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows tellurium decomposition into atomic, s -orbital and p -orbitals respectively.

The orbital decomposition is shown in the last two columns of Figure 9, where it indicates that the d -orbitals of manganese dominate slightly and there are no contributions from tellurium s -orbitals. Regarding MnI_2 , the projections do not change significantly with the inclusion of SOC, the results can be seen in Appendix A.2.

4.3 Al_2VSe_4

For the final material, we select the last metal listed in Table 1, Al_2VSe_4 . As previously, we examine the effect of SOC non-self-consistently, and find the result seen in Figure 10.

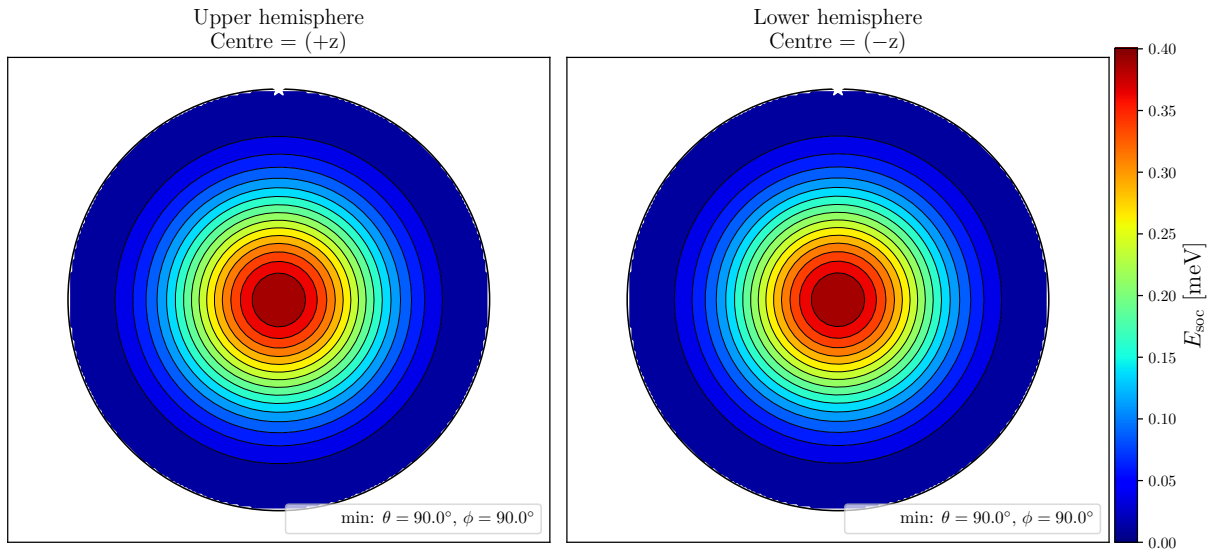


Figure 10: Energy cost of different orientations of $\hat{n}$ for Al_2VSe_4 , given in polar angle coordinates. The energy scale goes from the global minimum to the global maximum. The minimum is seen to be at the equator, and as expected, chirality plays no role.

After employing the method described in the previous sections, we obtain the band structure shown in Figure 11. Here, the left panel is without SOC and right panel is with SOC. In both figures, we see the odd-parity structure is strongest around the two parabolic bands closest to the Fermi level. In both situations, we have partially filled conduction bands crossing the Fermi level and, due to charge neutrality, we need partial empty valence bands in another region in the Brillouin zone.

The inclusion of SOC does not change the band structure drastically, one has to carefully investigate the structure to see the difference. On the first fully occupied band below the Fermi level, a clear avoided crossing is observed along the path between $-\bar{M}$ and Γ . The effect of SOC is smaller in this material compared to MnI_2 and MnTe_2 , as its constituent atoms are much lighter.

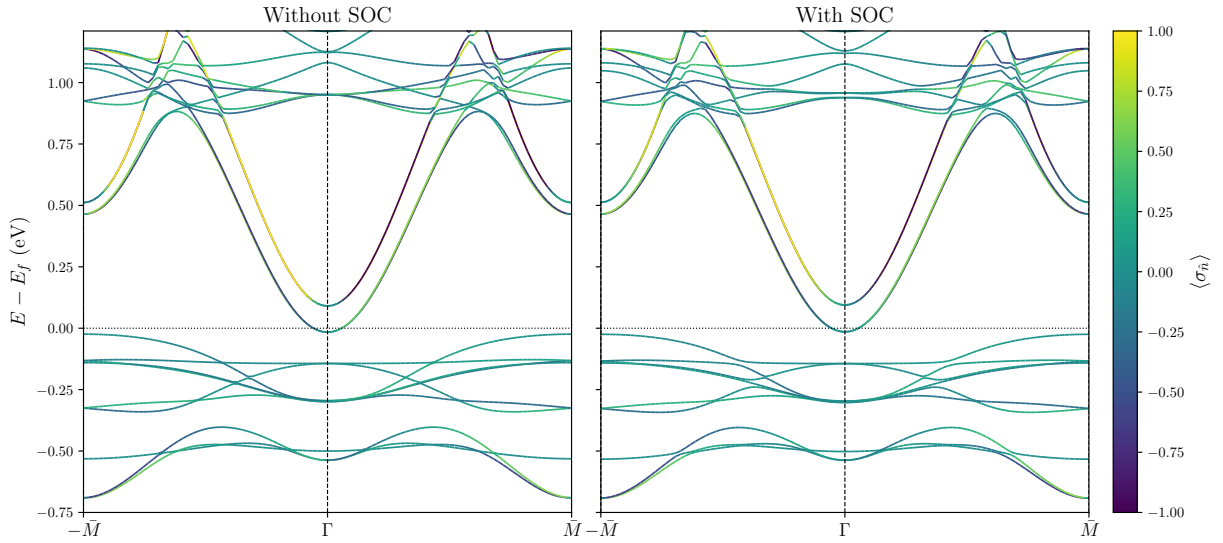


Figure 11: Spin-resolved band structure for Al_2VSe_4 for (left) without SOC and (right) with SOC and $\hat{n} = -\hat{z}$. The bands are coloured according to $\langle \sigma_n \rangle$. All dispersions are plotted along the path $-\bar{M} - \Gamma - \bar{M}$ in the magnetic Brillouin zone.

4.3.1 Atomic- and Orbital Projections

We have focused on the atomic position since there was no clear conclusion for the orbital decomposition. In Figure 12 we can see the atomic weights for (left) Al, (middle) Se and (right) V. For the electronic bands in the vicinity of the Fermi level, the states are predominantly derived from Al and Se, whereas V-derived states are dominant at energies below the Fermi level. The orbital decomposition is

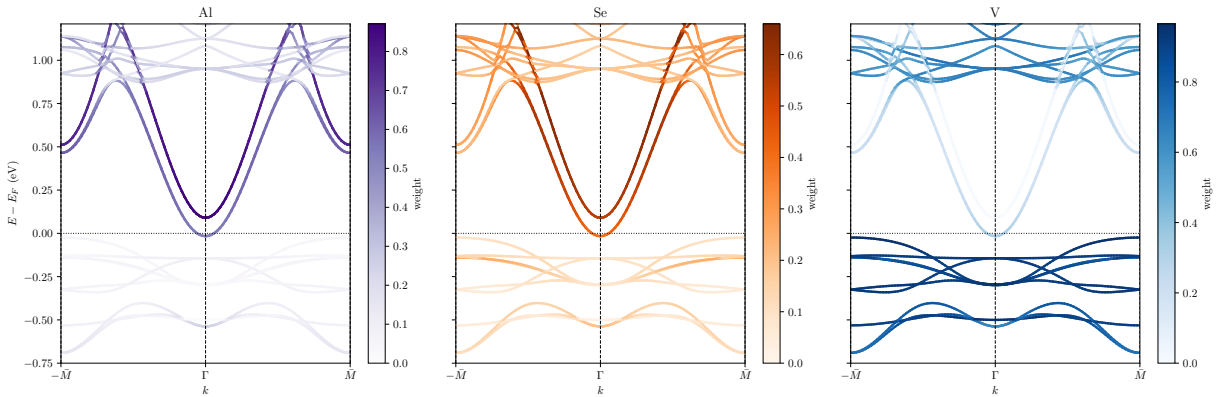


Figure 12: Atomic decomposition for Al_2VSe_4 without SOC. (left) Al, (middle) Se and (right) V.

more intricate, as both Al and Se exhibit significant hybridization between s - and p -orbitals, whereas V is dominated by d -orbital contributions in the bands located below the Fermi level. The corresponding figure is provided in Appendix A.3.

4.4 Material comparison

We have analysed the insulator MnI_2 and the two metals MnTe_2 and Al_2VSe_4 . The insulator was used first and foremost as a benchmark, since the band structure has been calculated before.

For MnI_2 , the odd-parity structure disappears almost entirely with the inclusion of SOC, whereas for MnTe_2 and Al_2VSe_4 , the odd-parity character is significantly less altered.

The odd-parity structure is not generally present in all bands, so to find general characteristics of the odd-parity bands, we attempted an atomic and orbital decomposition. SOC did not affect the decompositions for any of the materials notably. No general similarities in the decompositions were found across the materials. Further calculations were done (not shown) to examine each of the individual m -values for the p - and d -orbitals, but no general traits were found here.

Of the chosen materials, MnI_2 and Al_2VSe_4 do not appear to be of practical use from a spintronics point of view. MnI_2 is insulating and the odd-parity structure, which appears far below the valence band maximum, disappears almost entirely with SOC. Though Al_2VSe_4 is a metal, and the odd-parity structure appears at the small Fermi surface even with SOC, the nearly degenerate spin-bands precludes practical application. MnTe_2 appears most interesting for a spintronics application. Here, the odd-parity structure is pronounced even with SOC, and it is present in the bands close to the Fermi surface. The ground state appears to have $\hat{n} = \hat{z}$, and the material becomes a semiconductor rather than a metal in this configuration.

5 Conclusion

Previous classifications of odd-parity magnets seen in e.g. Refs. [3] and [14] are notably done in the absence of SOC, where an effective TRS is present. The main goal of the project was therefore to investigate how SOC may alter the odd-parity spin-texture. We focused our investigation on a selected set of materials exhibiting spin-spiral ordering with wave vector $\mathbf{Q} = [1/3, 1/3]$ on a hexagonal lattice, with in-plane and out-of-plane spiral configurations. By examining the effect of SOC, we find that the odd-parity classification becomes unclear. The effective TRS is broken by the SOC, and the spins may become non-coplanar, potentially obscuring the odd-parity spin texture of the bands.

The main finding of this project is the absence of any clear unifying factor determining whether the odd-parity spin texture is preserved upon inclusion of SOC. In the insulating material MnI_2 , which exhibits strong SOC, only weak vestiges of the odd-parity spin texture remains in certain regions of the band structure. In contrast, MnTe_2 , despite having SOC of comparable magnitude, retains most of the odd-parity character. Furthermore, Al_2VSe_4 , which has very weak SOC, retains virtually all of the odd-parity structure.

Interestingly, for MnI_2 , we find that, if $\hat{n}$ is in-plane, the odd-parity structure appears more robust when including SOC. We expect that, if $\hat{n}$ is in the plane of the 2D material, then the SOC-induced spin-canting aligns with the non-zero spin-expectation value caused by the odd-parity structure. The effect of SOC on the odd-parity structure may therefore be smaller.

For future avenues, it would be interesting to examine different lattices, $\mathbf{Q}$ -vectors and spin-spiral orientations to see if more general statements about the effect of SOC on the odd parity structure can be made.

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7 Appendix

A Orbital projections with SOC

In this appendix we show the orbital projections for our three candidate materials including SOC.

A.1 MnI_2

In Figure 13 the projections are done with $\hat{n} = -\hat{z}$.

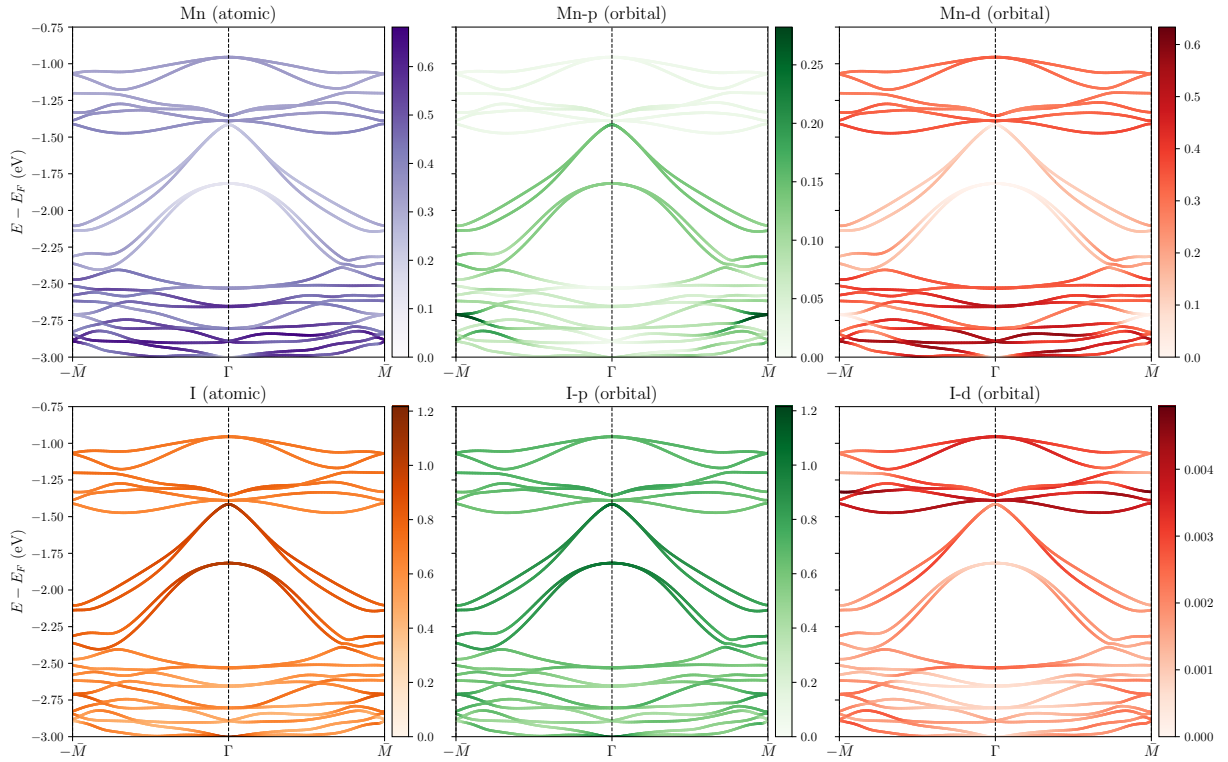


Figure 13: Results for MnI_2 with SOC and $\hat{n} = -\hat{z}$. The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows iodine decomposition into atomic, p -orbital and d -orbitals respectively. Note that the weights need not sum to 1.

In Figure 14 the projections are done with $\hat{n} = -\hat{x}$.

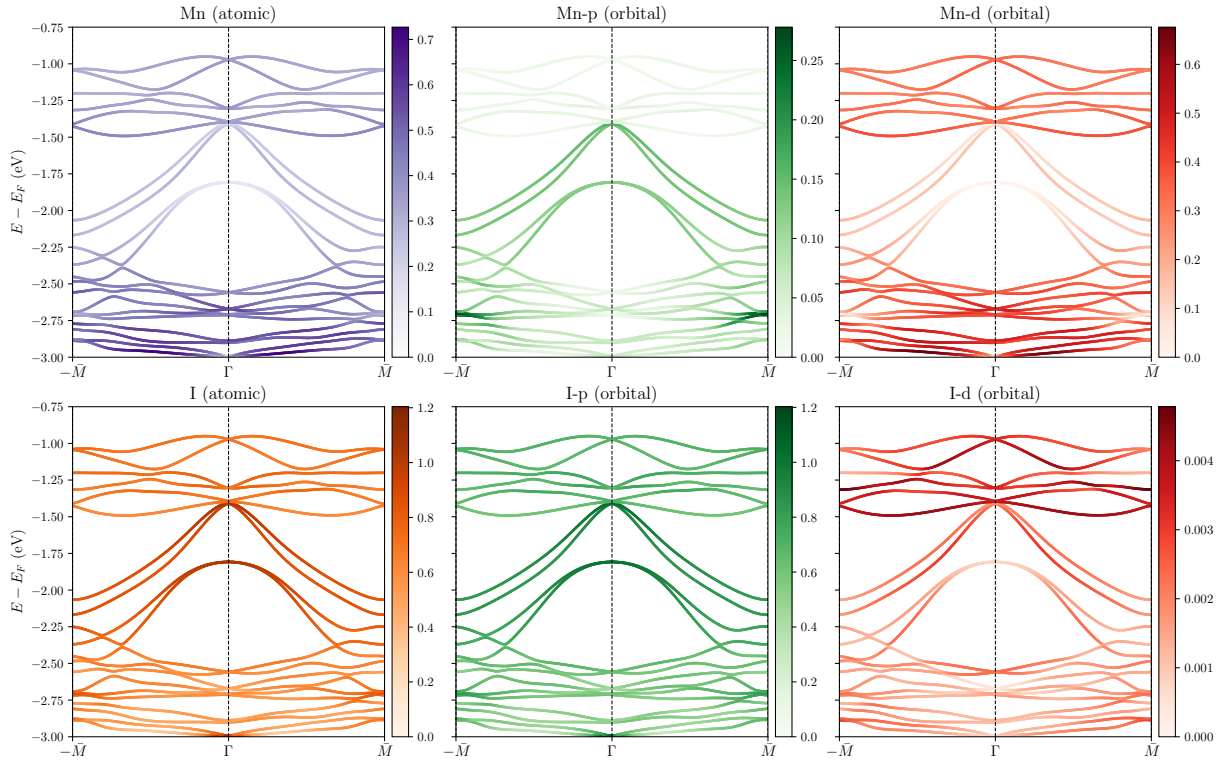


Figure 14: Results for MnI_2 with SOC and $\hat{n} = -\hat{x}$. The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows iodine decomposition into atomic, p -orbital and d -orbitals respectively. Note that the weights need not sum to 1.

A.2 MnTe_2

In Figure 15 the projections are done with $\hat{n} = \hat{z}$, which is the ground state.

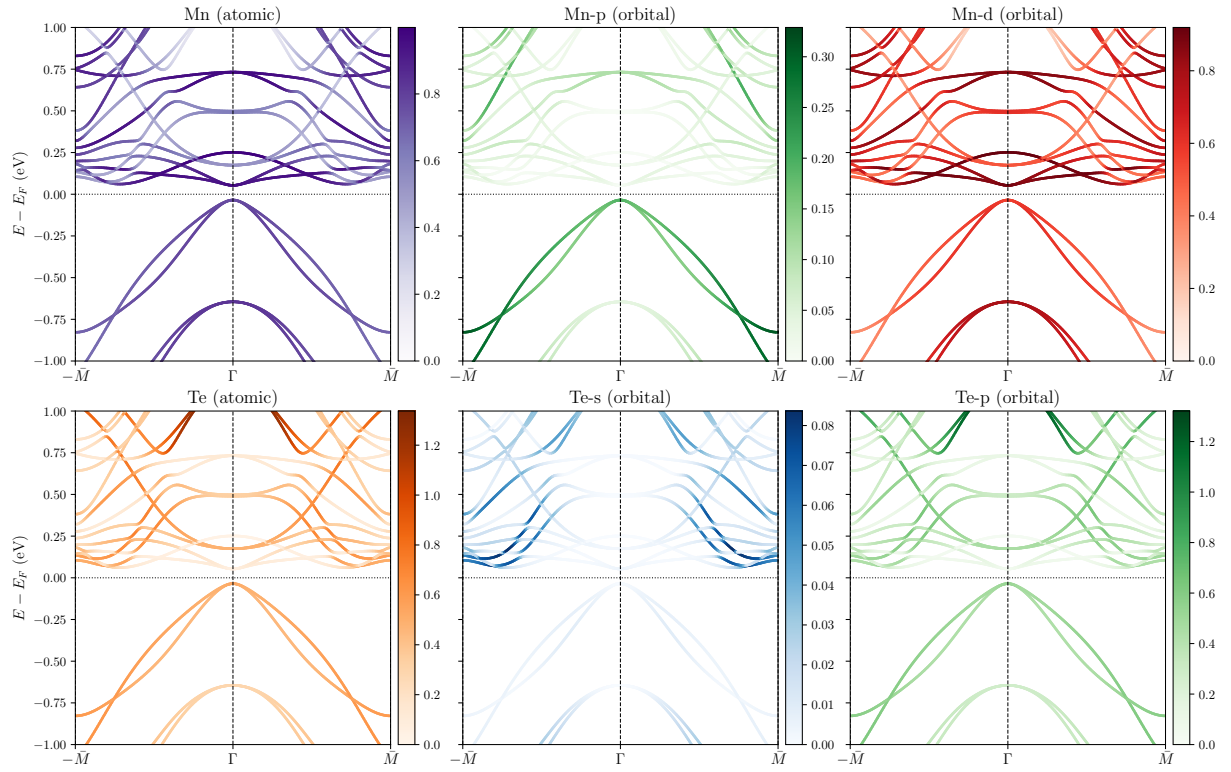


Figure 15: Results for MnTe_2 with SOC and $\hat{n} = \hat{z}$. The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows tellurium decomposition into atomic, p -orbital and d -orbitals respectively. Note that the weights need not sum to 1.

In Figure 16 the projections are done with $\hat{n} = -\hat{z}$, which is the opposite chirality as the ground state.

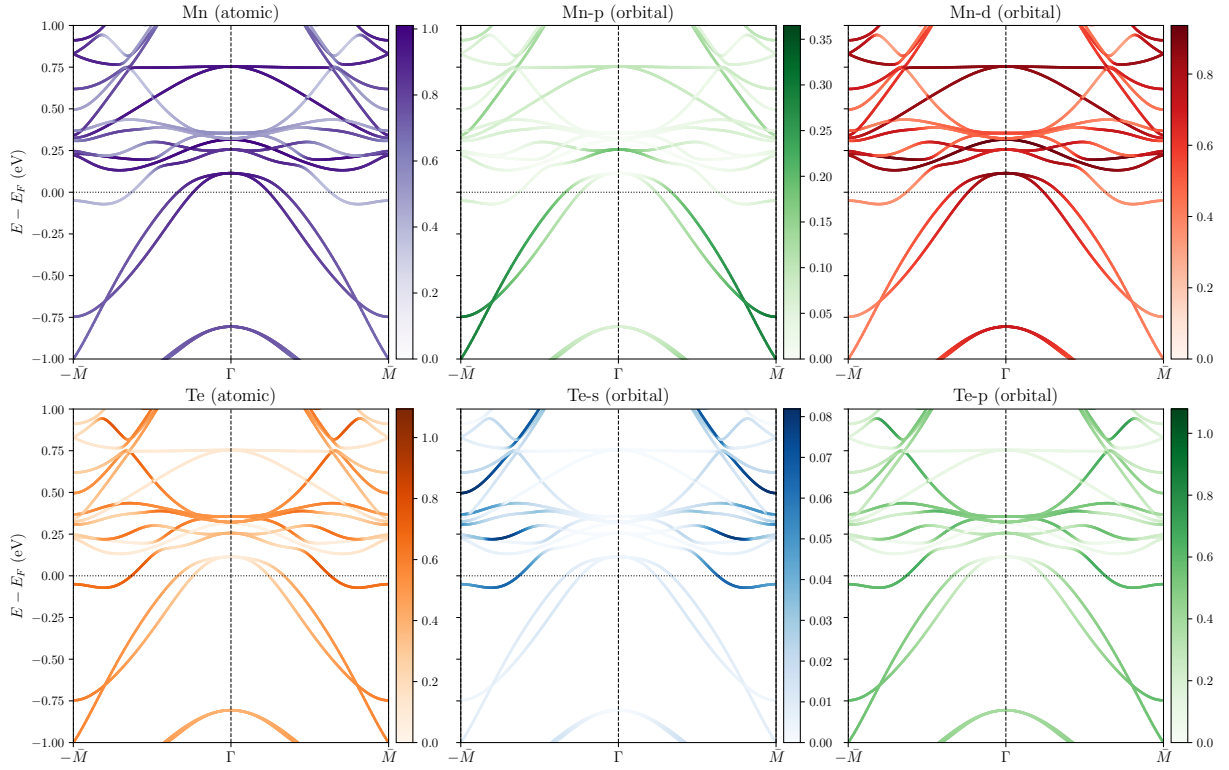


Figure 16: Results for MnTe_2 with SOC and $\hat{n} = -\hat{z}$. The upper row shows manganese decomposition into atomic, p -orbital and d -orbitals respectively. Likewise the lower row shows tellurium decomposition into atomic, p -orbital and d -orbitals respectively. Note that the weights need not sum to 1.

A.3 Al_2VSe_4

In Figure 17 the atomic projections are done with $\hat{n} = -\hat{x}$ and SOC.

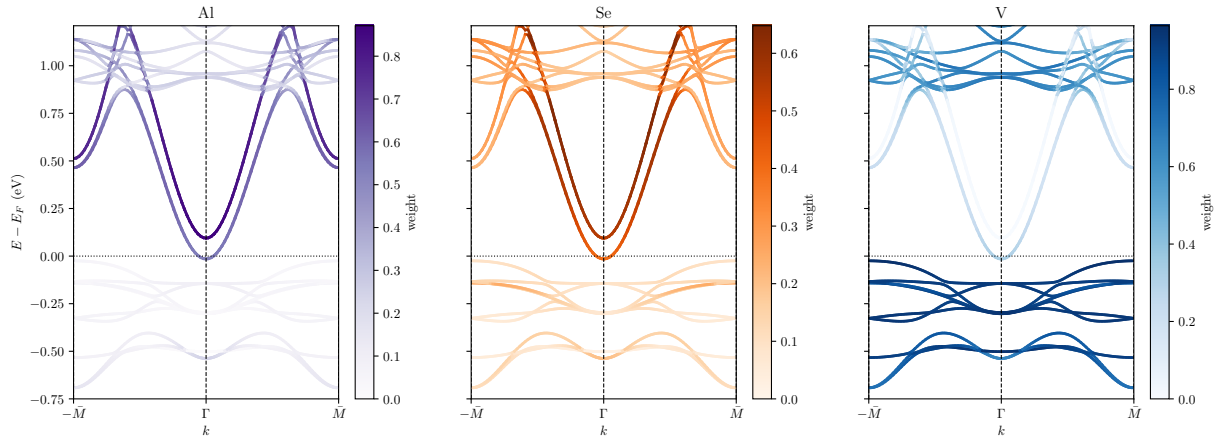


Figure 17: The atomic decomposition for Al_2VSe_4 with SOC and $\hat{n} = -\hat{x}$. Note that the weights need not sum to 1.

In Figure 18 the orbital projections are done with $\hat{n} = -\hat{x}$ and no SOC.

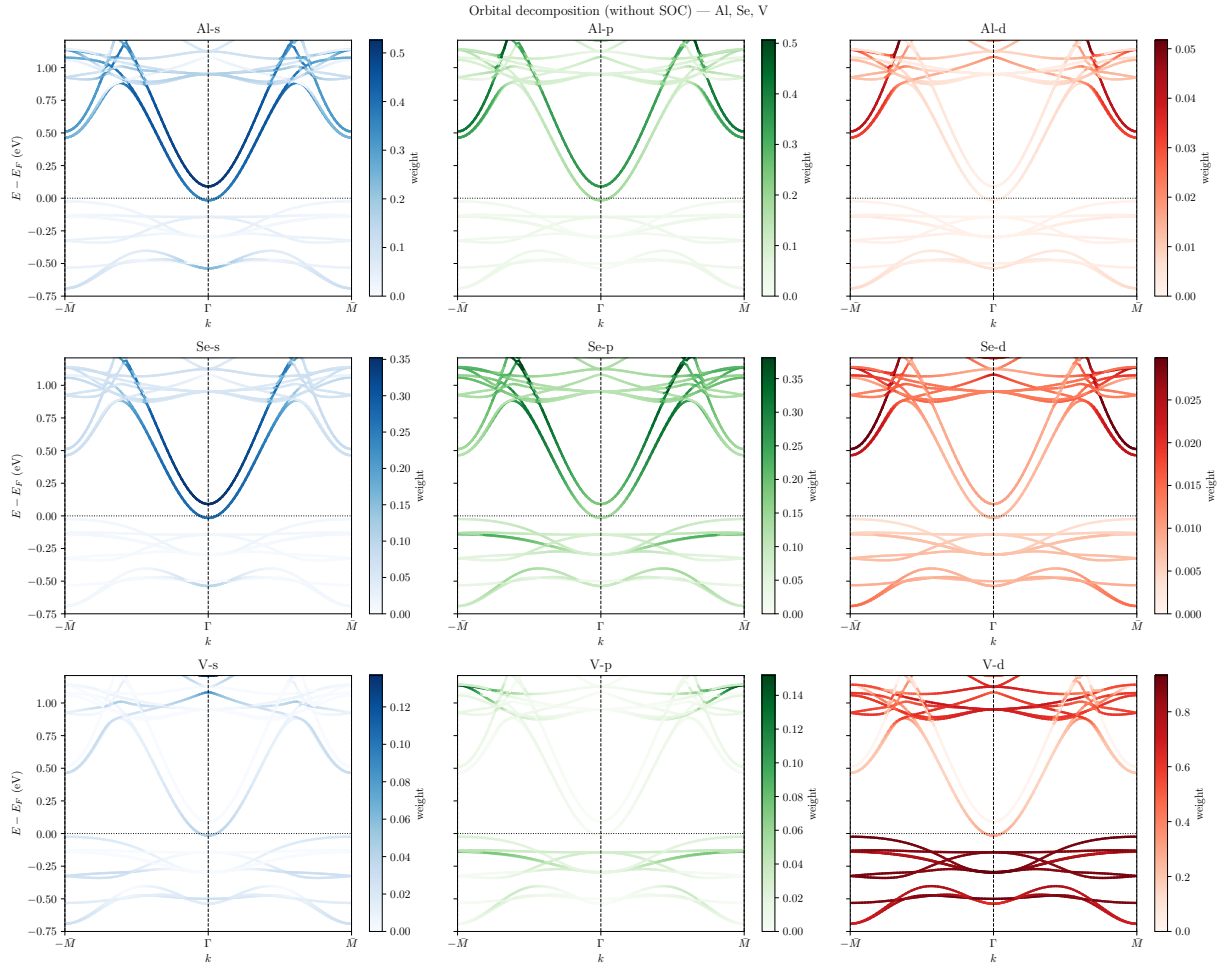


Figure 18: Orbital decomposition for Al_2VSe_4 without SOC and $\hat{n} = -\hat{x}$. The upper row shows aluminium decomposition into s -orbitals, p -orbitals and d -orbitals respectively. Likewise the middle row shows selenium decomposed into s -orbitals, p -orbitals and d -orbitals respectively. At last again the lower row shows vanadium decomposed into s -orbitals, p -orbitals and d -orbitals respectively. Note that the weights need not sum to 1.

In Figure 19 the orbital projections are done with $\hat{n} = -\hat{x}$ and SOC.

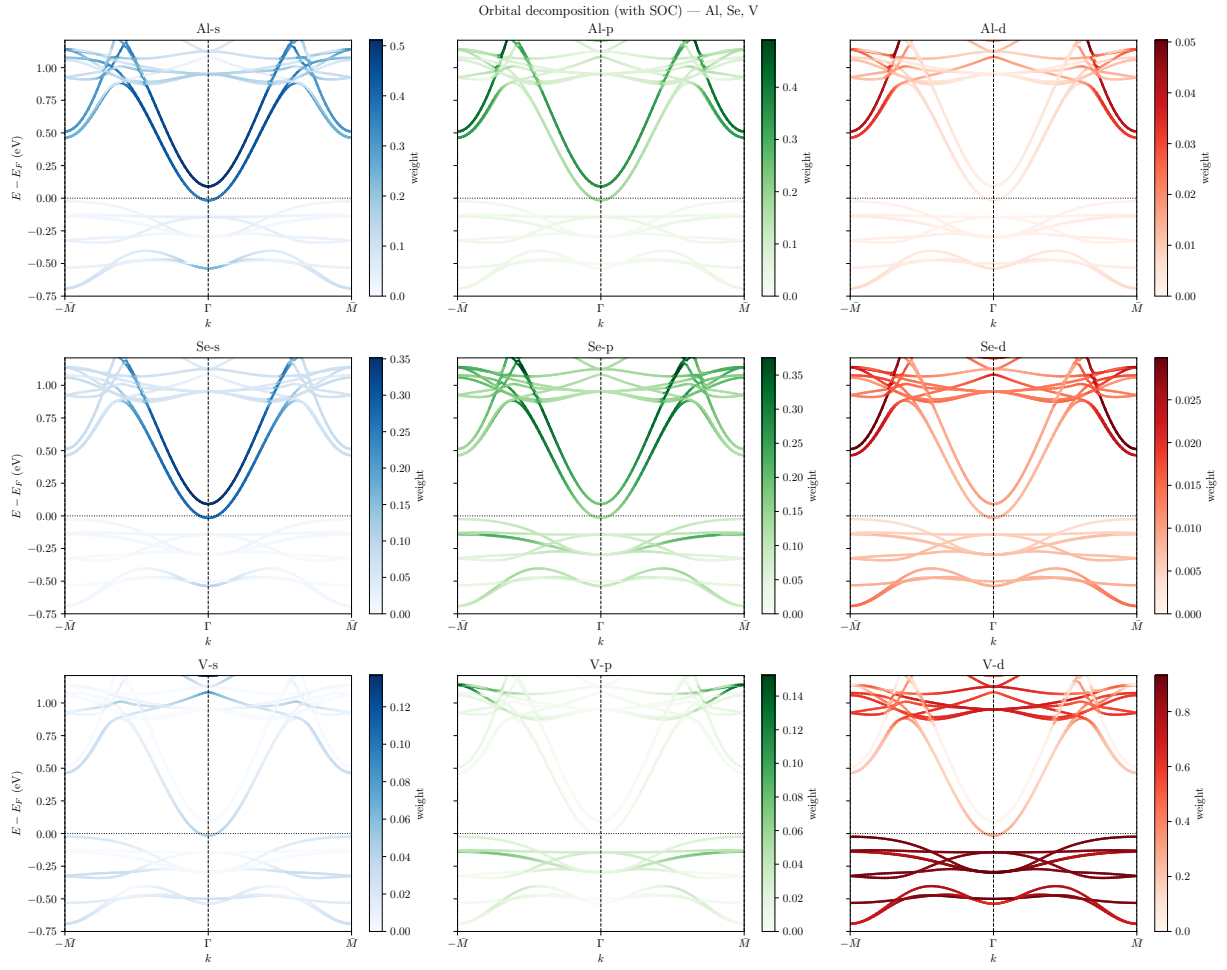


Figure 19: Orbital decomposition for Al_2VSe_4 with SOC and $\hat{n} = -\hat{x}$. The upper row shows aluminium decomposition into s -orbitals, p -orbitals and d -orbitals respectively. Likewise the middle row shows selenium decomposed into s -orbitals, p -orbitals and d -orbitals respectively. At last again the lower row shows vanadium decomposed into s -orbitals, p -orbitals and d -orbitals respectively. Note that the weights need not sum to 1.